

MATANZAS, Cuba

WMO: 782290

Lat: 23.034N

Lon: 81.435W

Elev: 210

StdP: 14.58

Time Zone: -5.00 (CUB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 52.5	(c) 55.7	(d) 46.0	(e) 46.3	(f) 58.3	(g) 48.4	(h) 50.7	(i) 61.3	(j) 21.1	(k) 72.2	(l) 19.0	(m) 72.2	(n) 3.8	(o) 160	(p) 0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 16.6	(c) 91.5	(d) 76.6	(e) 90.0	(f) 76.7	(g) 89.5	(h) 76.7	(i) 80.5	(j) 87.5	(k) 79.7	(l) 86.7	(m) 79.1	(n) 86.1	(o) 8.6	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	150.8	85.6	77.4	143.5	84.4	77.0	141.7	84.1	44.3	87.8	43.4	87.0	42.7	86.5	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.3	(b) 17.3	(c) 15.5	(e) 45.2	(f) 94.5	(g) 2.4	(h) 2.9	(i) 43.4	(j) 96.5	(k) 42.0	(l) 98.2	(m) 40.7	(n) 99.8	(o) 39.0	(p) 101.9		
(a) 19.3	(b) 17.3	(c) 15.5	(e) 44.3	(f) 82.8	(g) 2.3	(h) 2.7	(i) 42.7	(j) 84.7	(k) 41.4	(l) 86.3	(m) 40.1	(n) 87.9	(o) 38.5	(p) 89.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.3	70.2	71.5	73.0	76.3	78.4	80.1	80.7	81.0	80.2	77.9	73.7	72.2
	DBStd	5.20	4.96	5.10	4.95	3.66	2.68	1.99	1.64	1.77	1.74	2.88	3.69	4.63
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	41	17	11	5	0	0	0	0	0	0	0	1	7
	CDD50	9596	626	602	714	788	879	903	953	962	905	865	712	687
	CDD65	4162	178	193	254	338	414	453	488	497	455	400	263	229
	CDH74	33200	879	1232	1907	2702	3533	4006	4675	4643	3939	2987	1496	1201
	CDH80	11340	141	292	568	903	1271	1514	1906	1927	1493	858	247	220
(14) Wind	WSAvg	6.8	7.5	7.6	8.2	8.1	7.2	5.9	5.8	5.5	5.2	6.4	7.2	6.8
(15) Precipitation	PrecAvg	47.8	1.8	1.5	1.6	3.0	5.1	7.3	5.3	6.2	6.7	5.2	2.2	1.9
	PrecMax	60.3	6.8	4.8	3.9	5.9	15.7	15.9	8.3	11.3	13.0	14.4	6.0	6.4
	PrecMin	36.1	0.2	0.3	0.7	0.9	1.5	3.2	2.4	3.7	3.0	1.5	0.1	0.3
	PrecStd	6.3	1.6	1.2	0.8	1.3	2.9	3.0	1.4	1.6	2.0	2.9	1.5	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.0	88.2	91.1	91.8	92.8	91.7	91.7	93.0	91.5	89.7	86.4	87.4
		MCWB	71.4	71.2	72.7	73.5	75.2	77.5	77.5	78.8	77.6	76.4	75.0	73.6
	2%	DB	83.5	85.9	88.1	89.6	89.8	89.9	90.0	91.2	89.7	87.9	84.5	84.4
		MCWB	71.0	71.3	72.0	73.3	74.8	77.5	77.6	77.6	77.5	76.8	74.1	73.2
	5%	DB	81.0	83.9	85.9	87.7	88.0	88.7	89.5	89.8	88.2	86.2	82.7	82.5
		MCWB	70.8	71.5	71.7	73.0	74.5	77.4	77.7	77.8	77.5	76.6	73.5	72.3
	10%	DB	79.0	81.0	83.7	85.6	86.3	87.6	88.0	88.2	87.5	84.6	80.9	80.5
		MCWB	69.8	70.5	71.2	72.8	74.1	77.3	77.6	77.9	77.5	75.8	72.2	71.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.5	75.9	77.2	79.8	80.2	81.5	81.6	81.9	80.6	79.8	78.0	77.2
		MCDB	81.6	82.3	85.7	86.4	86.7	88.0	88.0	89.0	87.1	85.6	83.6	82.2
	2%	WB	73.4	74.3	75.2	76.4	78.2	79.7	80.0	80.1	79.5	78.7	76.3	74.9
		MCDB	79.2	81.2	83.3	84.1	85.4	86.7	87.3	87.3	86.3	84.8	82.0	80.8
	5%	WB	72.2	72.9	73.7	74.7	76.9	78.8	79.2	79.4	78.9	77.8	75.1	73.8
		MCDB	78.2	79.8	82.0	83.4	84.9	85.8	86.5	86.7	85.8	84.0	80.4	79.8
	10%	WB	71.0	71.8	72.4	73.7	75.6	77.9	78.5	78.7	78.1	76.8	74.0	72.6
		MCDB	77.0	78.6	80.5	82.1	83.5	85.1	85.9	86.1	85.1	83.0	79.0	78.4
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	18.2	19.1	18.4	17.2	15.6	16.3	16.6	15.7	14.2	14.7	15.7
		MCDBR	18.3	20.0	21.5	20.6	19.1	17.1	17.4	17.8	16.8	15.6	15.7	17.3
		MCWBR	8.8	8.7	8.7	8.0	7.8	8.0	8.1	8.1	8.0	7.3	7.8	8.3
	5% WB	MCDBR	16.3	17.8	19.0	18.2	16.6	15.9	16.2	16.6	15.8	14.5	13.4	15.4
		MCWBR	8.7	8.6	9.0	8.5	8.1	8.4	8.6	8.5	8.3	7.8	7.5	8.2
(40) Clear-Sky Solar Irradiance	taub		0.357	0.365	0.374	0.401	0.424	0.453	0.456	0.434	0.414	0.404	0.374	0.367
	taud		2.538	2.517	2.478	2.400	2.376	2.345	2.328	2.410	2.461	2.473	2.507	2.528
	Ebn at Noon		289	293	294	286	277	267	266	273	278	277	280	280
	Edh at Noon		30	32	35	39	39	40	41	38	35	33	30	29
(44) All-Sky Solar Radiation	RadAvg		1248	1527	1834	2043	1984	1886	2043	1938	1719	1476	1288	1176
	RadStd		95	92	69	62	198	132	101	75	89	100	56	88

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.65	+1.73	N/A	N/A	N/A	N/A	N/A	-20	N/A	N/A

Nomenclature: See separate page